Nmap Port Scanning Lab — Complete Report

Kali Linux and Docker Setup, TCP Port Scanning, SYN Scan, and Packet Trace

Date performed: 12 October 2026 | Evidence screenshots: 21

1. Objective

To set up Nmap in a Kali Linux Docker container, scan the authorized Nmap practice host, inspect selected TCP ports, troubleshoot a restricted-container raw-socket error, and perform a practice scan and packet trace against google.com as shown in the lab exercise.

2. Environment and Tools

Platform: Skills Network Cloud IDE. Environment: Kali Linux running in Docker. Tool: Nmap 7.991 (version shown in the screenshots). Targets used in the recorded exercise: scanme.nmap.org (Nmap's practice host) and google.com (public website used in the course practice exercise).

3. Environment Setup

The lab instructions guided the learner to obtain the provided Dockerfile, build a Kali Linux image, and launch an interactive container. The screenshots document the Docker build, container launch, and installation of Nmap. The course-provided download URL is not repeated here because it may be temporary or account-specific.

4. Nmap Installation

The following commands were run inside the Kali container:
apt-get update
apt-get install -y nmap
nmap --version

5. Scan of scanme.nmap.org

A basic scan was run against scanme.nmap.org, followed by a targeted scan of TCP ports 22, 113, and 139. In the recorded output, 22/tcp was open (ssh), while 113/tcp (ident) and 139/tcp (netbios-ssn) were closed. The output also reported many filtered TCP ports. These are time-specific observations and can change.

6. Raw-Socket Permission Troubleshooting

A SYN scan launched as an unprivileged user inside the restricted container failed with a raw-socket / Operation not permitted error. This is consistent with container permissions or missing network capabilities; it does not mean that the target itself is unreachable. The lab then used debug and packet-trace options to inspect Nmap's scan activity.

7. Practice Exercise: google.com

The practice exercise first ran a default scan against google.com. The captured output reported the host as up and showed 80/tcp open (http) and 443/tcp open (https). The targeted SYN scan against ports 80 and 443 reported the same states. DNS resolved the hostname to different IP addresses in separate runs, which is normal for large, load-balanced services.

8. Commands Executed

```
Environment setup:
docker build . -t kalilinux
docker run --tty --interactive kalilinux

Install Nmap:
apt-get update
apt-get install -y nmap

Practice-host scans:
nmap scanme.nmap.org
```

```
nmap -p 22,113,139 scanme.nmap.org
nmap -sS -p 22,113,139 scanme.nmap.org
nmap -d --packet-trace -p 22,113,139 scanme.nmap.org

Google practice scans:
nmap google.com
nmap -sS -p 80,443 google.com
nmap --packet-trace -p 80,443 google.com
```

9. Summary of Observed Results

The recorded scans identified 22/tcp open for SSH on scanme.nmap.org, with ports 113/tcp and 139/tcp closed. In the google.com practice exercise, ports 80/tcp (HTTP) and 443/tcp (HTTPS) were reported open. Packet-trace output for the Google scan showed outgoing TCP SYN packets and responses with SYN/ACK flags on the scanned ports.

Target	Port / State	Service
scanme.nmap.org	22/tcp — Open	SSH
scanme.nmap.org	113/tcp — Closed	Ident
scanme.nmap.org	139/tcp — Closed	NetBIOS-SSN
google.com	80/tcp — Open	HTTP
google.com	443/tcp — Open	HTTPS

10. Port-State Interpretation

Open: a service is accepting connections on that port. Closed: the host responded, but no service was listening on that port. Filtered: Nmap could not determine whether the port was open because filtering or lack of response prevented a definitive result. A port scan reveals network exposure; it does not, on its own, prove that a vulnerability exists.

11. Conclusion

This lab provided hands-on practice with Nmap installation, host discovery, targeted TCP port scanning, SYN-scan behavior, port-state interpretation, and packet-level troubleshooting. It also demonstrated that scan techniques may be limited by permissions in a containerized environment.

12. Limitations and Responsible Use

Results are snapshots and may vary with DNS resolution, load balancing, time, network path, and filtering. Scan only systems you own or have explicit authorization to test, and follow the target operator's published rules and course instructions. The use of a public practice target in a learning exercise is not a claim that the target is vulnerable.

13. Evidence Screenshots — Complete Lab

All screenshots provided for the lab are included below in the order of the setup, Nmap practice-host scan, and Google practice exercise.

Figure 1: 01 Course Lab Overview

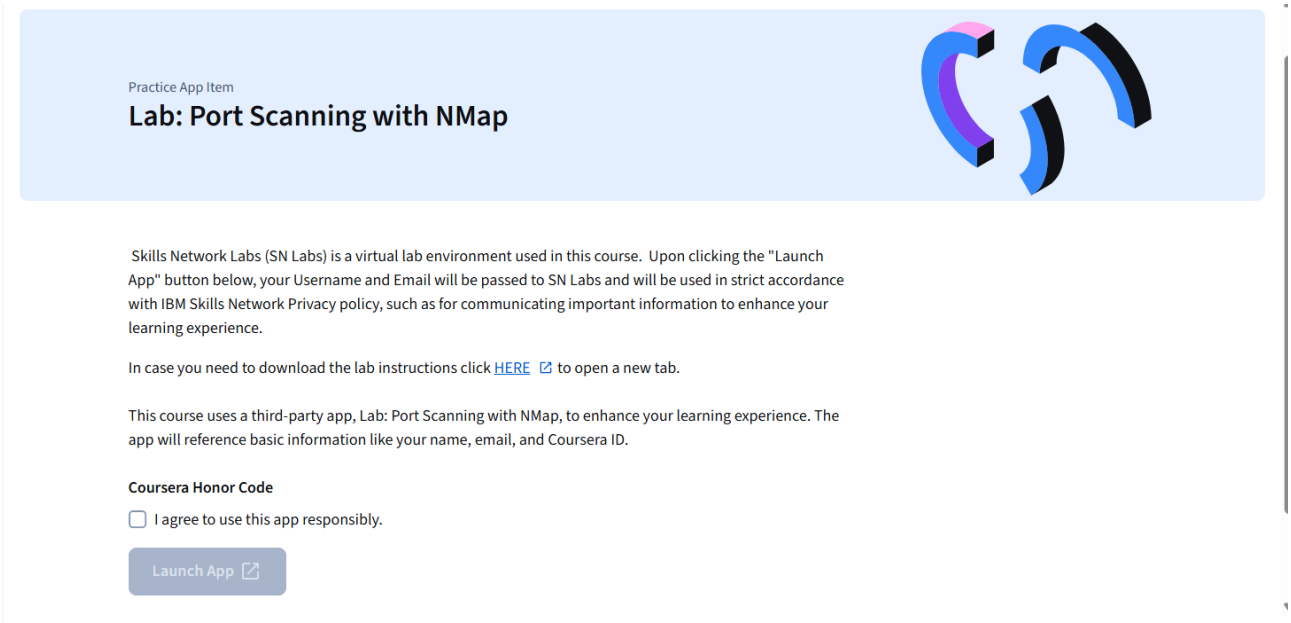


Figure 2: 02 Cloud Ide Setup

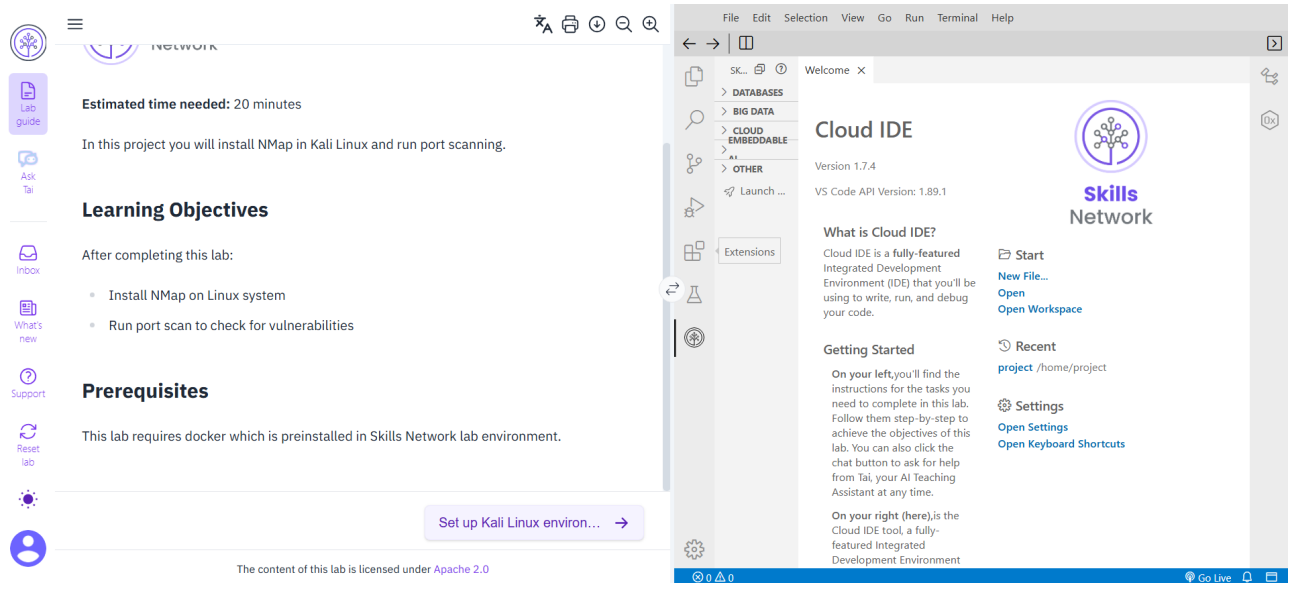


Figure 3: 03 Docker Image Download

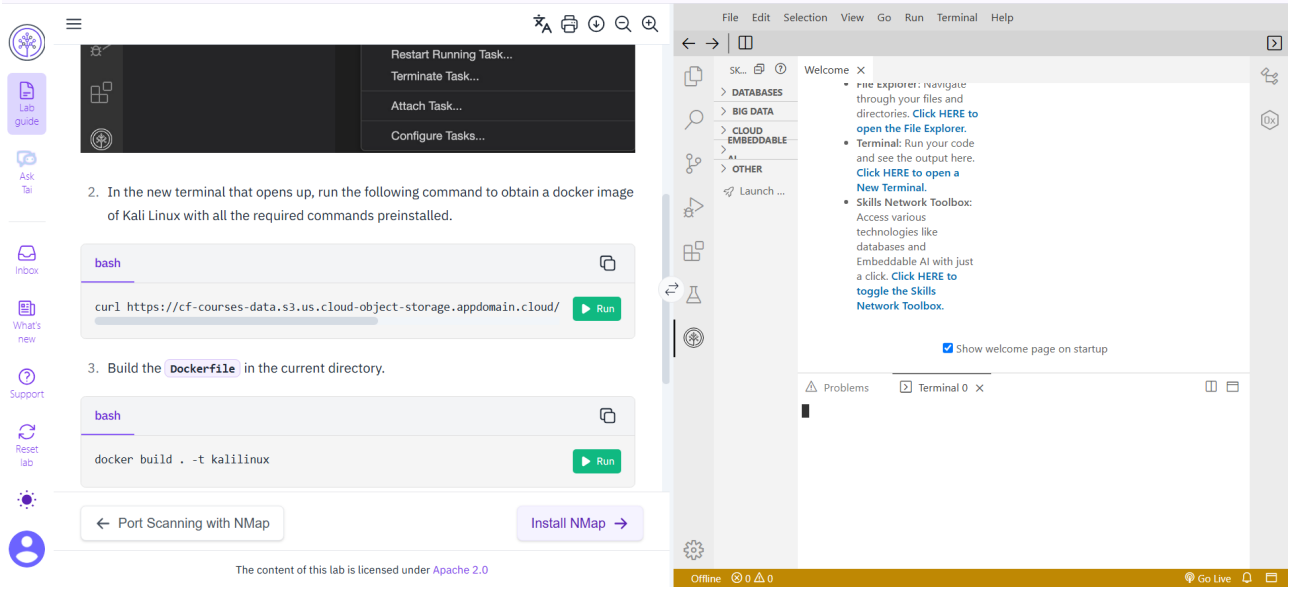


Figure 4: 04 Docker Build

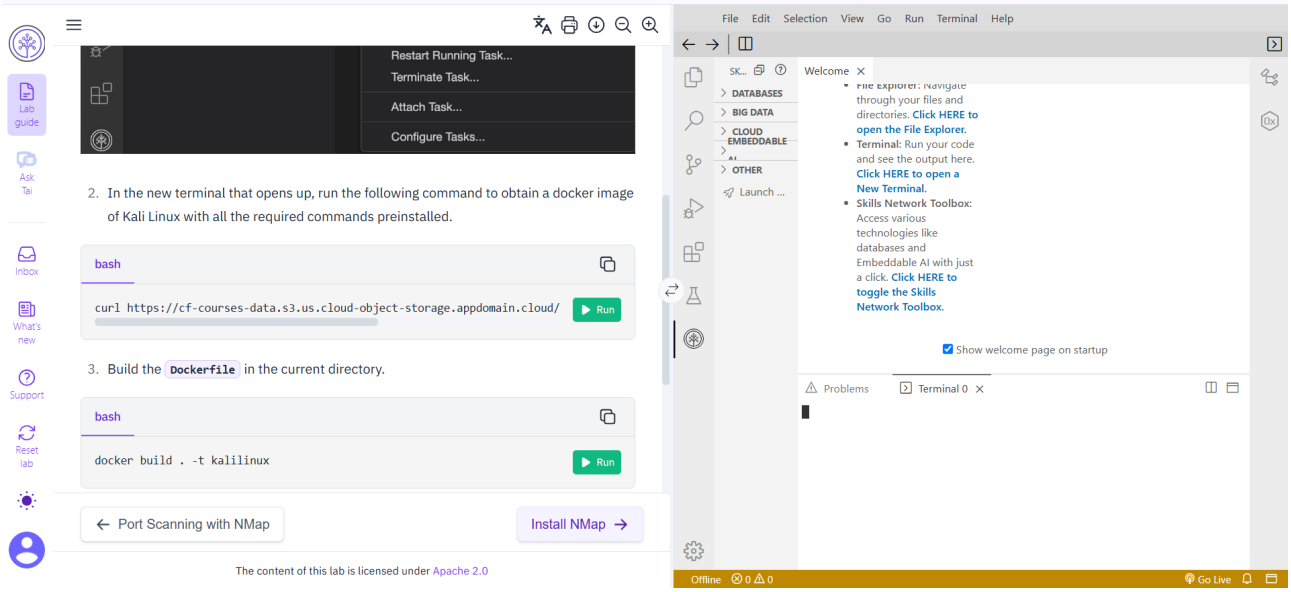


Figure 5: 05 Kali Container

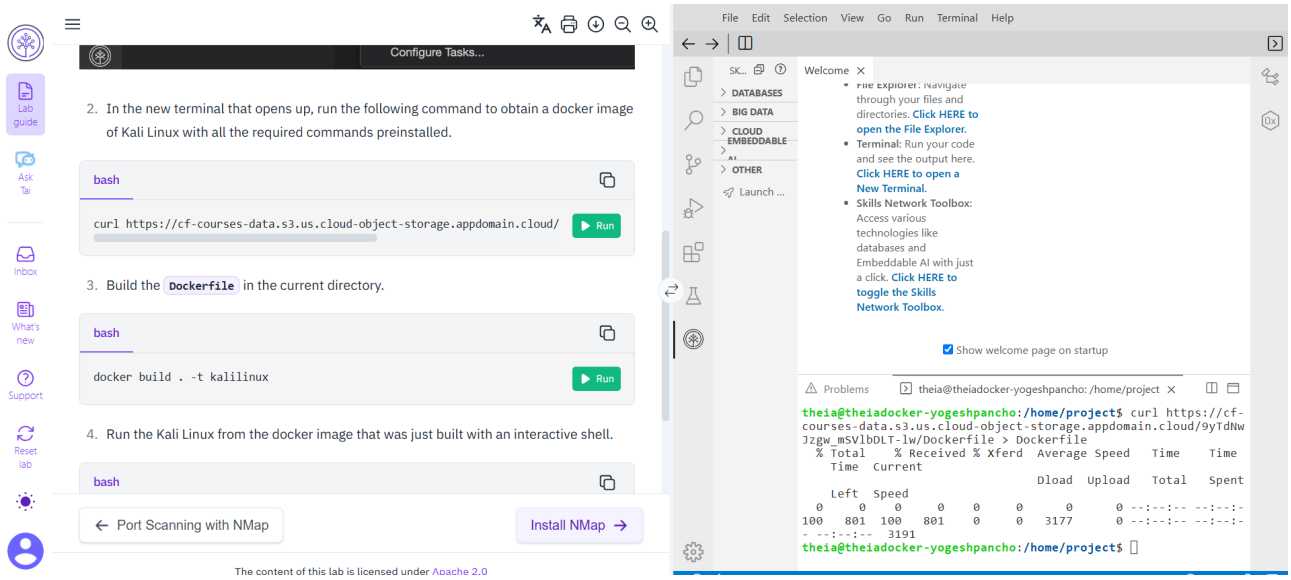


Figure 6: 06 Install Nmap

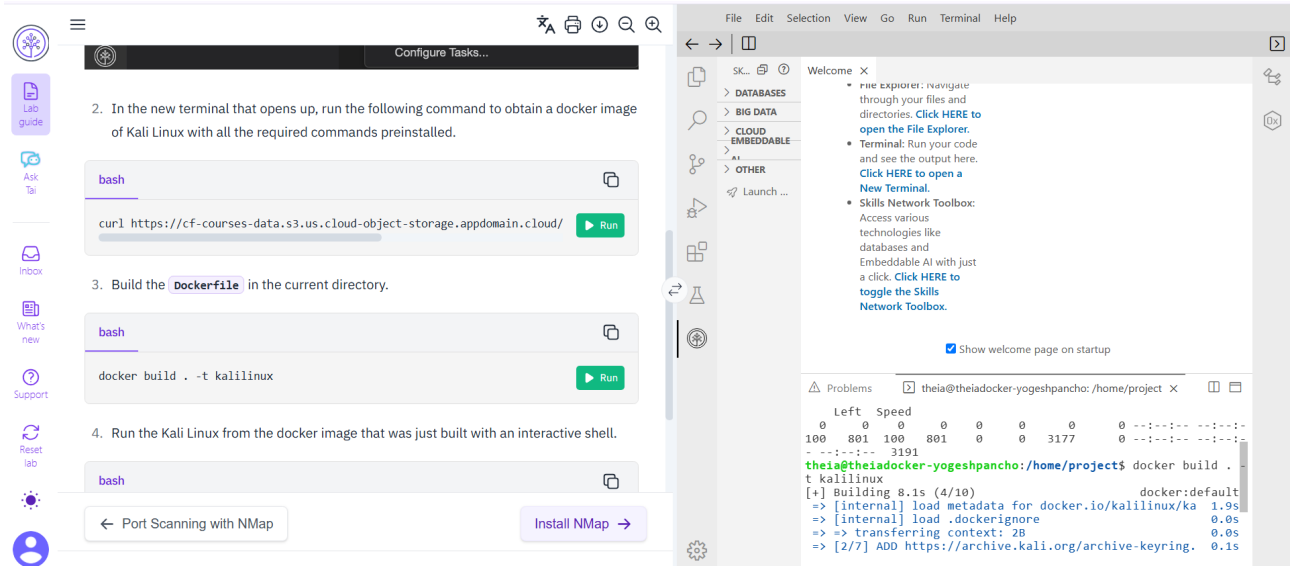


Figure 7: 07 Nmap Install Output

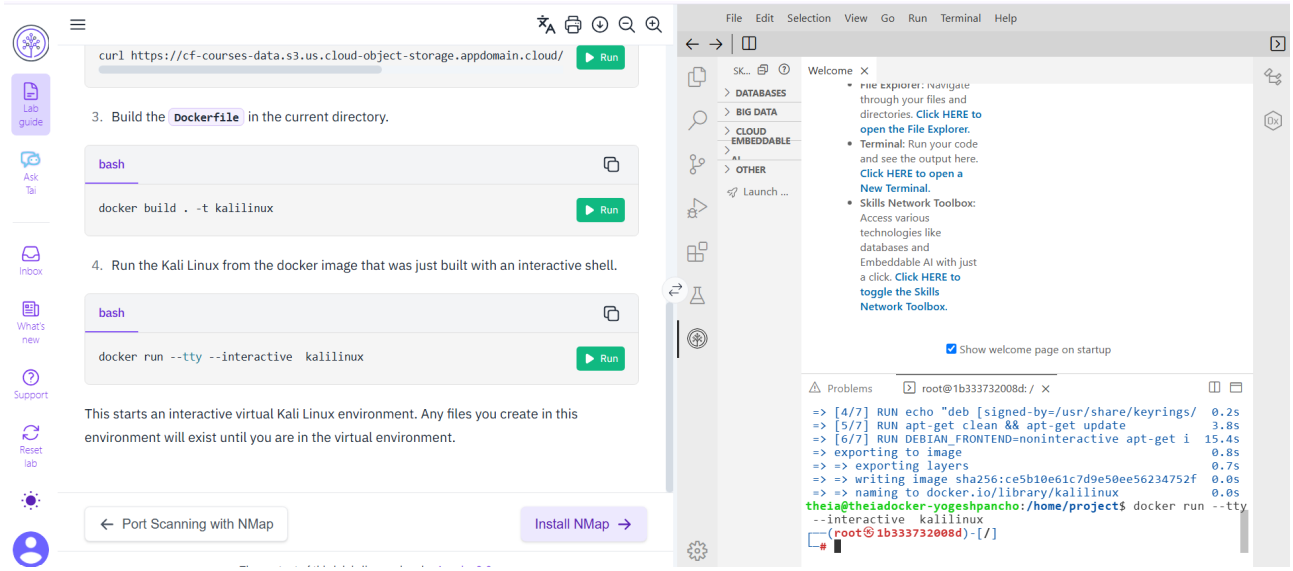


Figure 8: 08 Scan Scanme Basic

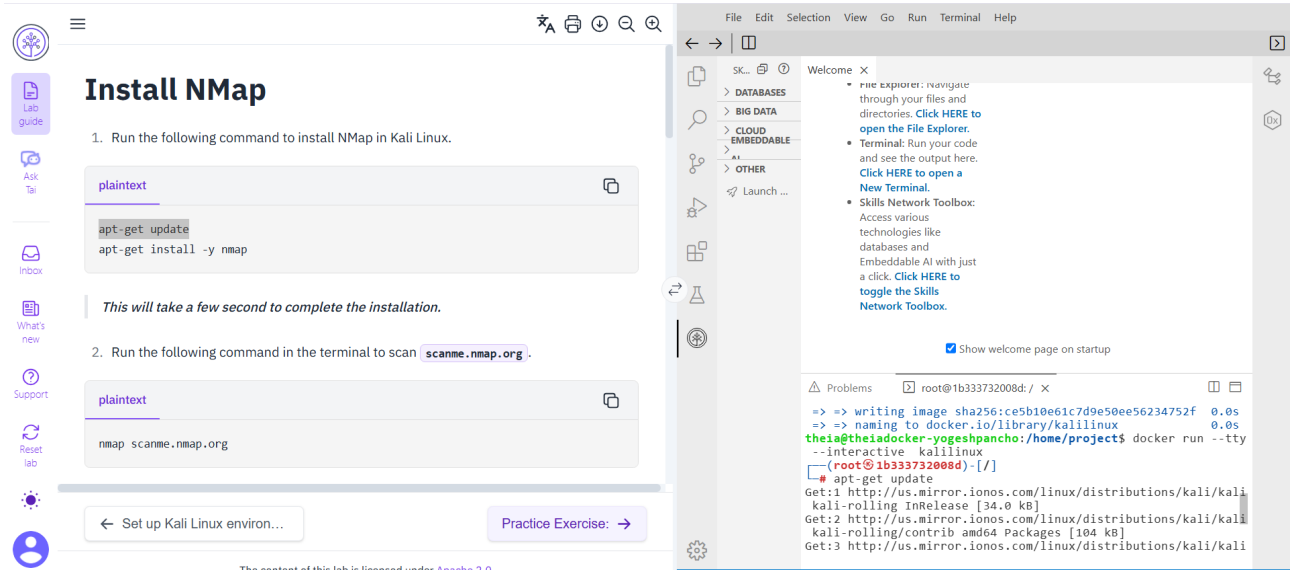


Figure 9: 09 Scan Scanme Selected Ports

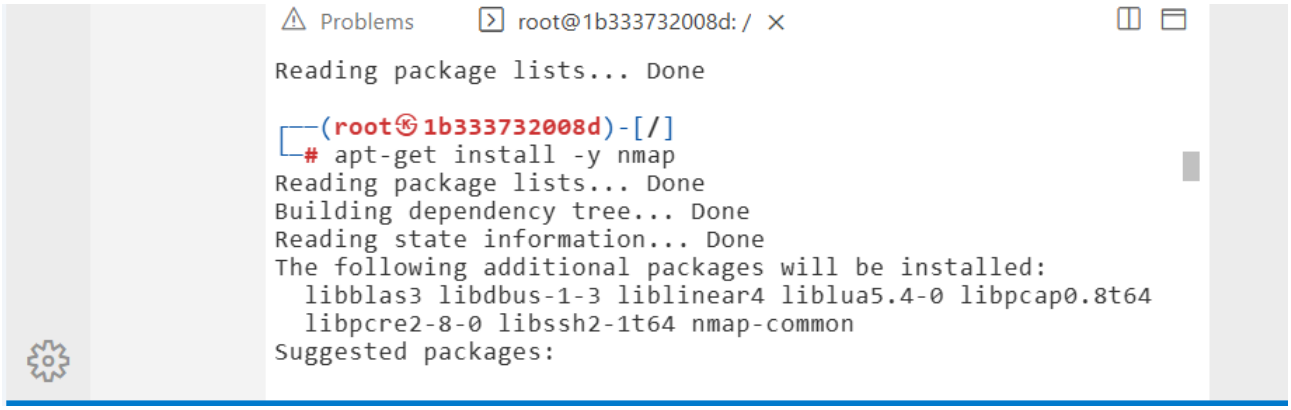


Figure 10: 10 Unprivileged Syn Scan Error

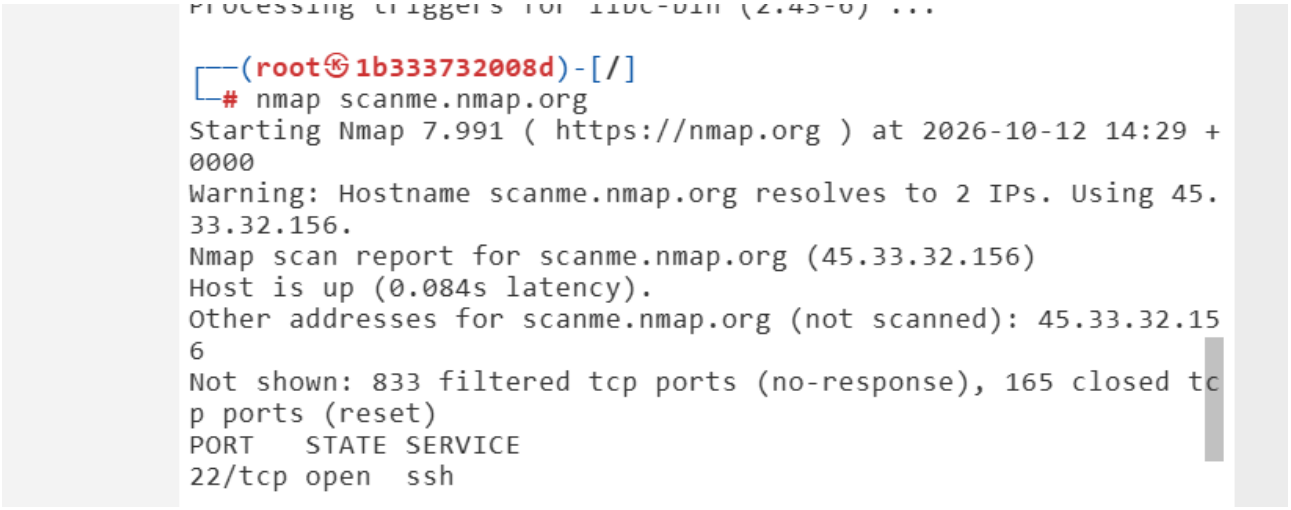


Figure 11: 11 Syn Scan Permission Troubleshooting

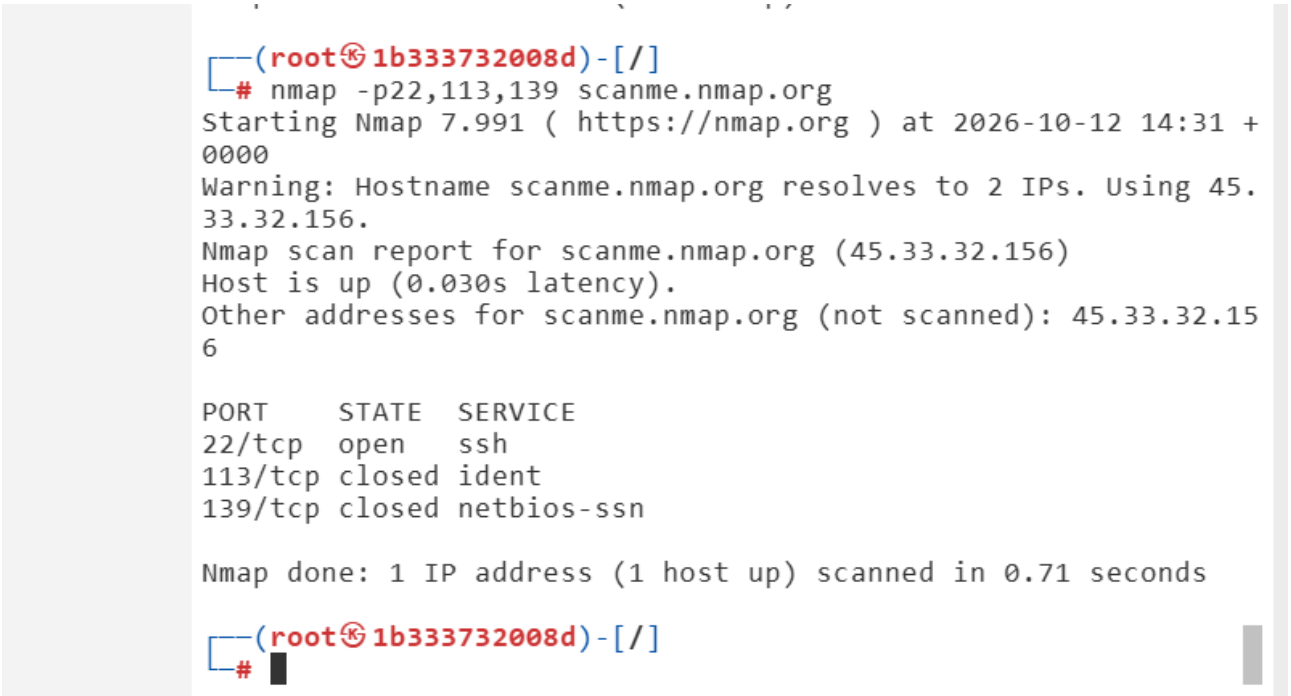


Figure 12: 12 Packet Trace Scanme

```
┌──(root㉿1b333732008d)-[/]
└─# useradd John

┌──(root㉿1b333732008d)-[/]
└─# su John
$ nmap -p22,113,139 scanme.nmap.org
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-12 14:32 +
0000
Warning: Hostname scanme.nmap.org resolves to 2 IPs. Using 45.
33.32.156.
Couldn't open a raw socket. Error: (1) Operation not permitted
dnet: failed to open device eth0
Couldn't open a raw socket or eth handle.
QUITTING!
$
```

Figure 13: 13 Debug Packet Trace

```
┌──(root㉿1b333732008d)-[/]
└─# su John
$ nmap -p22,113,139 scanme.nmap.org
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-12 14:32 +
0000
Warning: Hostname scanme.nmap.org resolves to 2 IPs. Using 45.
33.32.156.
Couldn't open a raw socket. Error: (1) Operation not permitted
dnet: failed to open device eth0
Couldn't open a raw socket or eth handle.
QUITTING!
$ nmap -p22,113,139 -sS scanme.nmap.org
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-12 14:32 +
0000
Warning: Hostname scanme.nmap.org resolves to 2 IPs. Using 45.
33.32.156.
Couldn't open a raw socket. Error: (1) Operation not permitted
dnet: failed to open device eth0
Couldn't open a raw socket or eth handle.
QUITTING!
$ exit

┌──(root㉿1b333732008d)-[/]
└─#
```

Figure 14: 14 Port State Reasons

```
┌──(root㉿1b333732008d)-[/]
└─# nmap -d --packet-trace -p22,113,139 scanme.nmap.org
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-12 14:33 +0000
--------------- Timing report ---------------
  hostgroups: min 1, max 100000
  rtt-timeouts: init 1000, min 100, max 10000
  max-scan-delay: TCP 1000, UDP 1000, SCTP 1000
  parallelism: min 0, max 0
  max-retries: 10, host-timeout: 0
  min-rate: 0, max-rate: 0
---------------------------------------------
mass_dns: Using DNS server 192.168.65.1
Initiating Parallel DNS resolution of 1 host. at 14:33
NSOCK INFO [0.0290s] nsock_iod_new2(): nsock_iod_new (IOD #1)
NSOCK INFO [0.0290s] nsock_connect_udp(): UDP connection requested to 1
92.168.65.1:53 (IOD #1) EID 8
NSOCK INFO [0.0290s] nsock_trace_handler_callback(): Callback: CONNECT
SUCCESS for EID 8 [192.168.65.1:53]
NSOCK INFO [0.0290s] nsock_read(): Read request from IOD #1 [192.168.65
.1:53] (timeout: -1ms) EID 18
NSOCK INFO [0.0290s] nsock_write(): Write request for 33 bytes to IOD #
1 EID 27 [192.168.65.1:53]
NSOCK INFO [0.0290s] nsock_write(): Write request for 33 bytes to IOD #
1 EID 35 [192.168.65.1:53]
NSOCK INFO [0.0290s] nsock_trace_handler_callback(): Callback: WRITE SU
CCESS for EID 35 [192.168.65.1:53]
```

Figure 15: 15 Final Scanme Output

```
┌──(root㉿1b333732008d)-[/]
└─# nmap -d5 --packet-trace -p22,113,139 scanme.nmap.org
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-12 14:36 +0000
Fetchfile found /usr/share/nmap/nmap-services
Fetchfile found /usr/share/nmap/nmap-protocols
Fetchfile found /usr/share/nmap/nmap.xsl
The max # of sockets we are using is: 0
--------------- Timing report ---------------
  hostgroups: min 1, max 100000
  rtt-timeouts: init 1000, min 100, max 10000
  max-scan-delay: TCP 1000, UDP 1000, SCTP 1000
  parallelism: min 0, max 0
  max-retries: 10, host-timeout: 0
  min-rate: 0, max-rate: 0
---------------------------------------------
mass_dns: Using DNS server 192.168.65.1
NSOCK DEBUG [-814409.7640s] nsock_set_loglevel(): Set log level to DEBU
G
Initiating Parallel DNS resolution of 1 host. at 14:36
NSOCK INFO [0.0300s] nsock_iod_new2(): nsock_iod_new (IOD #1)
NSOCK DEBUG [0.0300s] event_new(): event_new (IOD #1) (EID #8)
NSOCK INFO [0.0300s] nsock_connect_udp(): UDP connection requested to 1
```

Figure 16: 16 Lab Terminal Evidence

```
Scanned at 2026-10-12 14:36:12 UTC for 0s

PORT    STATE  SERVICE     REASON
22/tcp  open   ssh         syn-ack ttl 37
113/tcp closed ident       reset ttl 37
139/tcp closed netbios-ssn reset ttl 37
Final times for host: srtt: 30717 rttvar: 38454  to: 184533

Read from /usr/share/nmap: nmap-protocols nmap-services.
Nmap done: 1 IP address (1 host up) scanned in 0.69 seconds
           Raw packets sent: 7 (284B) | Rcvd: 4 (164B)
```

Figure 17: 17 Google Basic Scan

Practice Exercise:

1. Try to nmap one of the public websites. eg., `google.com`.
2. Make a note of the open ports. Run nmap SYN scan on the open ports.
3. Do a `packet-trace` to the same ports that you have scanned above.

Figure 18: 18 Google Syn Scan

```
┌──(root㉿1b333732008d)-[/]
└─# nmap google.com
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-12 14:40 +0000
Warning: Hostname google.com resolves to 2 IPs. Using 142.251.218.78.
Nmap scan report for google.com (142.251.218.78)
Host is up (0.040s latency).
Other addresses for google.com (not scanned): 142.251.218.78
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 11.48 seconds

┌──(root㉿1b333732008d)-[/]
└─#
```

Figure 19: 19 Google Packet Trace Start

```
┌──(root㉿1b333732008d)-[/]
└─# nmap -sS -p 80,443 google.com
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-12 14:48 +0000
Warning: Hostname google.com resolves to 2 IPs. Using 142.251.219.142.
Nmap scan report for google.com (142.251.219.142)
Host is up (0.019s latency).
Other addresses for google.com (not scanned): 142.251.219.142

PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 1.17 seconds

┌──(root㉿1b333732008d)-[/]
└─#
```

Figure 20: 20 Google Packet Trace Details

```
┌──(root㉿1b333732008d)-[/]
└─# nmap --packet-trace -p 80,443 google.com
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-12 14:49 +0000
NSOCK INFO [0.0260s] nsock_iod_new2(): nsock_iod_new (IOD #1)
NSOCK INFO [0.0260s] nsock_connect_udp(): UDP connection requested to 1
92.168.65.1:53 (IOD #1) EID 8
NSOCK INFO [0.0260s] nsock_trace_handler_callback(): Callback: CONNECT
SUCCESS for EID 8 [192.168.65.1:53]
NSOCK INFO [0.0260s] nsock_read(): Read request from IOD #1 [192.168.65
.1:53] (timeout: -1ms) EID 18
NSOCK INFO [0.0260s] nsock_write(): Write request for 28 bytes to IOD #
1 EID 27 [192.168.65.1:53]
NSOCK INFO [0.0260s] nsock_write(): Write request for 28 bytes to IOD #
```

Figure 21: 21 Google Packet Trace Summary

```
NSOCK INFO [0.0810s] nsock_read(): Read request from IOD #1 [192.168.65
.1:53] (timeout: -1ms) EID 34
NSOCK INFO [0.5830s] nsock_iod_delete(): nsock_iod_delete (IOD #1)
NSOCK INFO [0.5830s] nevent_delete(): nevent_delete on event #34 (type
READ)
SENT (0.5922s) TCP [172.17.0.2:35129 > 142.251.219.142:80 S seq=8628911
94 <mss 1460,wscale 5>] IP [ttl=52 id=29185 iplen=44 ]
SENT (0.5923s) TCP [172.17.0.2:35129 > 142.251.219.142:443 S seq=862891
194 <mss 1460,wscale 5>] IP [ttl=48 id=44653 iplen=44 ]
RCVD (0.6607s) TCP [142.251.219.142:443 > 172.17.0.2:35129 SA seq=22447
37882 ack=862891195 <mss 1460,wscale 5>] IP [ttl=37 id=0 iplen=44 ]
RCVD (0.6609s) TCP [142.251.219.142:80 > 172.17.0.2:35129 SA seq=411725
5599 ack=862891195 <mss 1460,wscale 5>] IP [ttl=37 id=0 iplen=44 ]
Nmap scan report for google.com (142.251.219.142)
Host is up (0.019s latency).
Other addresses for google.com (not scanned): 142.251.219.142

PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 0.68 seconds

┌──(root㉿1b333732008d)-[/]
└─#
```